

Work distribution for Assignment (Proposal)

Section	Task Distribution
Abstract & Area of study & Keywords	Member 1
Chapter 1 – Problem Statement and Motivation	Member 2
Chapter 1 – Scope, Objectives	Member 3
Chapter 1 – Contribution	Member 4
Chapter 1 – Report Organization	Member 4
Chapter 2 – Literature review 1	Member 1
Chapter 2 – Literature review 2	Member 2
Chapter 2 – Literature review 3	Member 3
Chapter 2 – Literature review 4	Member 4
Chapter 2 Comparison of the 4 techniques	Each student contributes here
Chapter 3 – System Design / Overview	Member 1
Chapter 3 – System Requirement (Hardware and software to use)	Member 2
Chapter 3 – Implementation Issues and Challenges	Member 3
Chapter 3 – Timeline & milestones for 13 weeks	Member 4
Chapter 4 – Coding work 1 & results	Member 1
Chapter 4 – Coding work 2 & results	Member 2
Chapter 4 – Coding work 3 & results	Member 3
Chapter 4 – Coding work 4 & results	Member 4
Chapter 5 – Conclusion	One paragraph Member 1 One paragraph Member 2 One paragraph Member 3 One paragraph Member 4
References (in IEEE style)	Each student contributes here
Appendices (e.g. list sample code/source code by each member)	Each student contributes here

UCCC2513 Assignment

A R&D proposal to detect traffic signs

The task here is to create a Research & Development (R&D) proposal to detect traffic signs reliably for autonomous vehicles. Students should work in groups of 4 members. Eighty-four test images are provided in the directory “Traffic signs” (which can be found in the directory of “Inputs” at MS Teams) for testing purpose. Figure 1 below shows one example of the test image and the desirable intermediate result to segment a sign. Figure 2 shows more examples of the signs with the colors red, blue and yellow. The shapes of the signs can be circle, triangle, rectangle and octagon.



Figure 1: One example of a test image and its segmentation result



Figure 2: Examples of 6 test images with different colors and shape

1. DETAILS

Your team or company needs funding to support this development and your career. You know of one mini-project fund of RM1,000,000 to support such development. Unfortunately, there are other teams / companies competing to get

this project and fund. Hence, your team needs to produce an exceptionally good R&D proposal to beat others.

After brainstorming, your team concludes that the combination of the information like color and shape can help to segment traffic signs reliably. Afterward, the task of detection/recognition would be easy. Hence, members should check on past work on effective color/shape detection and traffic sign recognition in general. To beat other teams, members should research the most promising and recent techniques. Afterward, the members can get together to share their findings to come up with a superior system design.

Finally, each team needs to submit a proposal (equivalent to the FYP1 report) to the funding reviewing committee to get the award of RM 1,000,000. To beat others, the proposal should highlight members' knowledge and expertise on the most updated techniques, e.g. color and shape detections (in chapter 2 of the proposal as shown in the next section below) and the superior system design with novel / new idea (in chapter 3). To play safe, some members should also look beyond color and shape into alternative approaches for sign segmentation / detection. The findings should be put in the proposal (chapter 2) too. The most updated and effective techniques should be within the last 10 (or 5) years since older approaches are likely obsolete.

Afterward, to show proof the proposed exciting design can work, some pieces of preliminary work should be carried out to clear doubt on the novel idea. This should be reported in chapter 4 of the proposal. Your proposal can then stand out tall to support application of the funding.

For this report, you need to present a complete design of the whole system. For chapter 4, the preliminary work should give result up to the part of sign segmentation as in chapter 1.

2. **REPORT**

The proposal report writing becomes easy by following the template file "FYP1_Report_Template_CS" from FICT FYP portal. The proposal will be marked according to the FYP1 Marking Rubrics. Five main chapters should be produced. They are summarized below.

Chapter	Contents	Marks
1	Introduction	3
2	Literature Review	12
3	Proposed Method / Approach	8
4	Preliminary Work	5
5	Conclusion	2
6	Presentation & demo	10

The content detail of each chapter will be discussed in class. Other small but important details that you should know are:

- a. **W⁵H**: This represents the terms What, Why, When, Who, Where and How with respect to the assignment title. Usually, we pay more attention to what, why and how when writing each chapter. In this report, you need to provide and explain what the system design is (i.e. the system block diagram or flowchart), why this design and how each of the details works together to form the final system.
- b. One **picture** is worth 1000 words. Hence, it will be to your advantage to provide and explain using system block diagrams or flowcharts.
- c. Excluding the code, diagrams and images from the intermediate results, the written report should not exceed **30 pages** with double spacing and font size 12 using the font “Times New Roman”.
- d. Report will be graded on **clarity** and **authenticity**.
- e. Each report section and source code section should be **tagged with name** of the group member who produces the writing and / or the implementation.
- f. University regulation prohibits **plagiarism**. If caught, you will end up with an “**F**” grade.
- g. Late project **will not be** accepted after the due date under any circumstances.
- h. Each team member is required to choose and develop one specific module as listed below for chapter 4 (Preliminary Work):

Member 1	Red sign segmentation using color information
Member 2	Blue sign segmentation using color information
Member 3	Yellow sign segmentation using color information
Member 4	Shape detection of signs to support the segmentation

- i. Google Form Submission Link:
https://docs.google.com/forms/d/e/1FAIpQLSdHFCcXRuUQNf343n-eRRXY5_eptHPTHNiC0aMB2PAIFY7w/viewform?usp=publish-editor
- j. Submission deadline: **27th July 2026 by 12pm**

3. **PRESENTATION & DEMONSTRATION**

Besides the report, each group is required to provide a **short video** of presentation of the proposal and demonstration running your code and displaying the preliminary work results. The demo should read the names of input image files from a specific directory.

Presentation Requirements:

1. The total presentation and Q&A session for each group shall not exceed **30 minutes**.
2. The recorded short video (proposal presentation, code demonstration, and preliminary results) shall be **a maximum of 25 minutes** in duration + followed by a **physical Q&A session of 5 minutes**.
3. The video quality must be clear and suitable for viewing.
4. Audio quality must be clear, audible, and free from significant background noise.
5. All group members are required to present their respective components in the short video presentation and participate in the physical Q&A session.

6. Throughout the recorded short video presentation, **all presenters are required to keep their cameras turned on** with their faces and student ID clearly visible on screen.

4. **HINTS**

One can find numerous pieces of demo and code using OpenCV from the Internet to segment traffic sign. You can refer / **reuse them for your code**. **Cite the sources** in your report if you reuse or adapt their code.